

Phase 2: Exploratory Data Analysis

StreamFlix Content Analytics Project

Name: Mark Maxwell Louis

In this notebook I make the 10 charts asked for in the brief and write a short observation under each one. I'm using the cleaned CSV files I saved at the end of Phase 1.

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt

subscribers = pd.read_csv('subscribers_clean.csv')
titles = pd.read_csv('titles_clean.csv')
watch_history = pd.read_csv('watch_history_clean.csv')
reviews = pd.read_csv('reviews_clean.csv')

subscribers['signup_date'] = pd.to_datetime(subscribers['signup_date'])
watch_history['watch_date'] = pd.to_datetime(watch_history['watch_date'])

print(watch_history.shape)
print(subscribers.shape)
```

```
(650000, 10)
(15000, 14)
```

I'll need genre, type, and country info from `titles.csv` matched up with each viewing session, so I'll merge `watch_history` and `titles` together on `title_id`. I'll use this merged table for a few of the charts below.

In [2]:

```
wh_titles = watch_history.merge(titles, on='title_id', how='left')
print(wh_titles.shape)
wh_titles.head()
```

```
(650000, 30)
```

Out[2]:

	watch_id	subscriber_id	title_id	watch_date	device	region	content_duration_min_x	watch_duration_min
0	1	SUB107630	TTL205207	2024-01-01	Smart TV	East Asia	888	826.7
1	2	SUB114773	TTL204889	2025-10-18	Mobile	South Asia	2340	1866.4
2	3	SUB106013	TTL206882	2026-04-26	Mobile	Latin America	656	75.0
3	4	SUB100939	TTL208849	2026-03-09	Mobile	North America	500	493.2
4	5	SUB107223	TTL202529	2023-12-27	Tablet	North America	141	94.6

5 rows × 30 columns

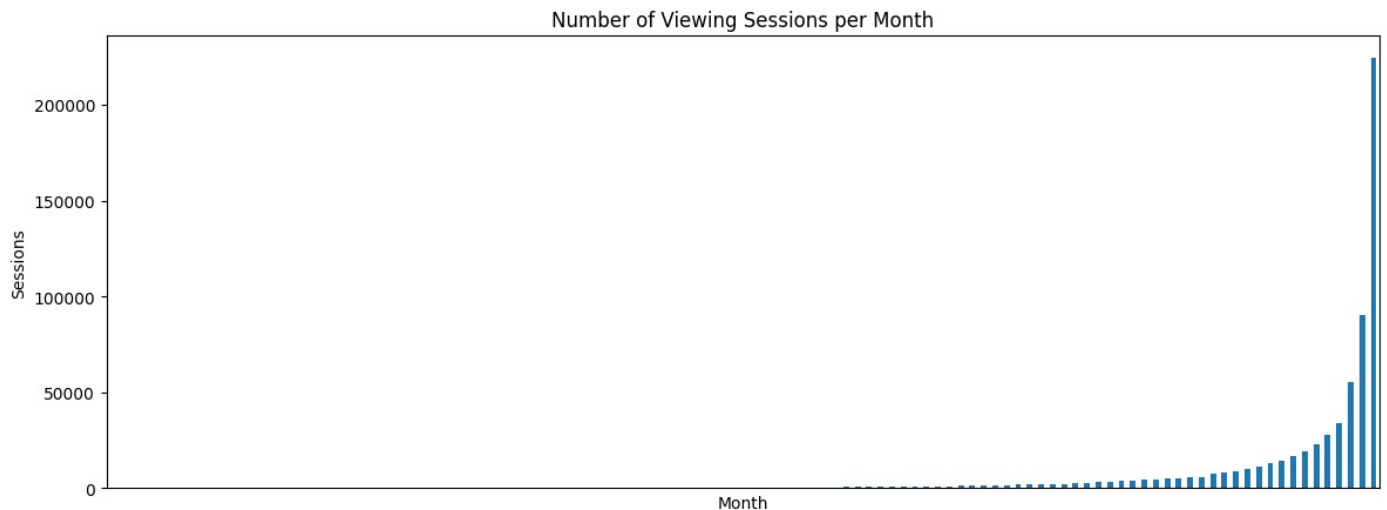


Chart 1: Monthly viewing volume (number of sessions per month)

In [3]:

```
watch_history['year_month'] = watch_history['watch_date'].dt.to_period('M')
monthly_sessions = watch_history.groupby('year_month').size()
monthly_sessions = monthly_sessions.sort_index()

monthly_sessions.plot(kind='bar', figsize=(14,5))
plt.title('Number of Viewing Sessions per Month')
plt.ylabel('Sessions')
plt.xlabel('Month')
plt.xticks([]) # too many months to label them all, so I'm hiding the labels
plt.show()
```



In [4]:

```
print('Top 5 busiest months:')
print(monthly_sessions.sort_values(ascending=False).head())
```

Top 5 busiest months:

year_month	
2026-05	224696
2026-04	90383
2026-03	55665
2026-02	33795
2026-01	27970

Freq: M, dtype: int64

Observation: Session volume keeps climbing from 2017 all the way to the most recent months, which makes sense since the subscriber base has been growing the whole time. The busiest months are all near the end of the timeline.

Chart 2: Monthly watch hours trend, compared year over year

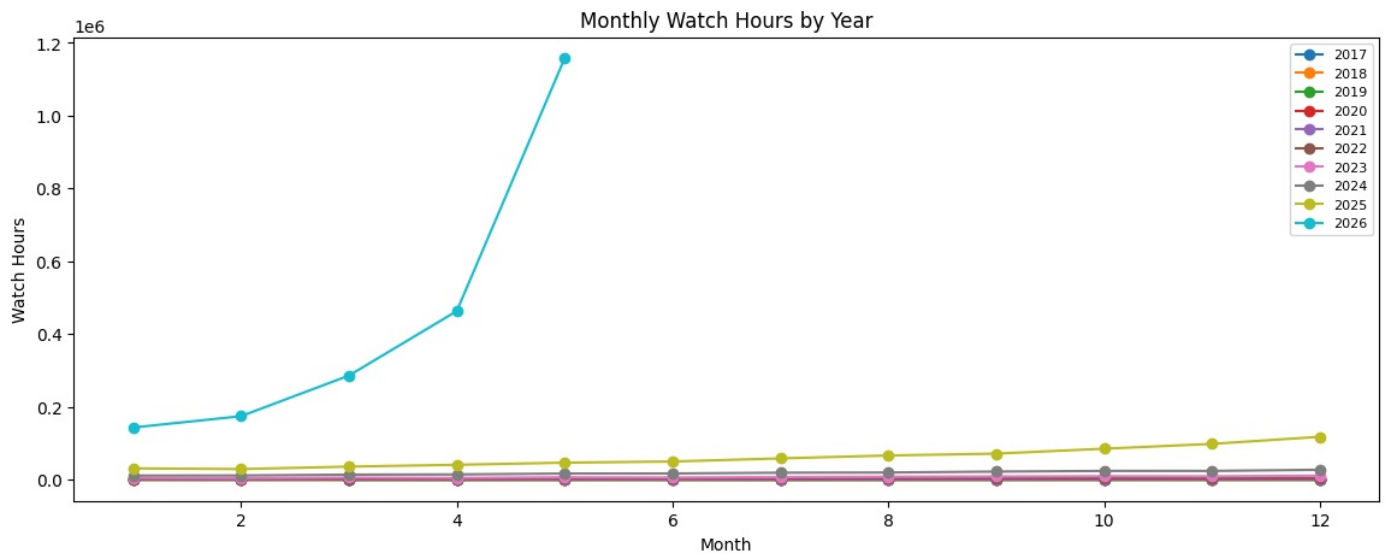
In [5]:

```
watch_history['hours'] = watch_history['watch_duration_min'] / 60
monthly_hours = watch_history.groupby('year_month')['hours'].sum()
monthly_hours = monthly_hours.sort_index()

# split the period into year and month so I can compare years on the same chart
monthly_hours_df = monthly_hours.reset_index()
monthly_hours_df.columns = ['year_month', 'hours']
monthly_hours_df['year'] = monthly_hours_df['year_month'].dt.year
monthly_hours_df['month'] = monthly_hours_df['year_month'].dt.month

plt.figure(figsize=(14,5))
for year in sorted(monthly_hours_df['year'].unique()):
    year_data = monthly_hours_df[monthly_hours_df['year'] == year]
    plt.plot(year_data['month'], year_data['hours'], marker='o', label=str(year))

plt.title('Monthly Watch Hours by Year')
plt.xlabel('Month')
plt.ylabel('Watch Hours')
plt.legend(fontsize=8)
plt.show()
```



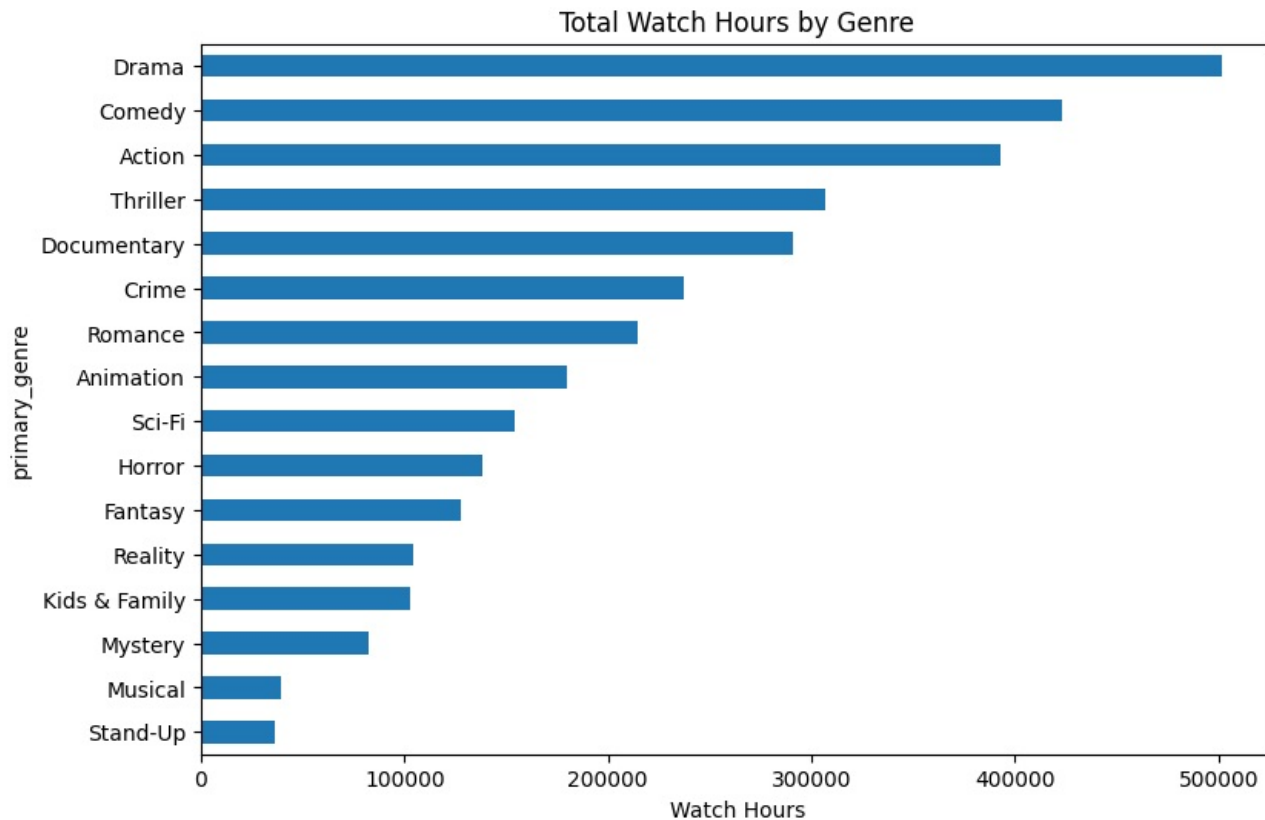
Observation: Each year's line sits above the year before it for most months. This tells me watch hours are growing faster than just the subscriber count, meaning people are also watching more per month than they used to.

Chart 3: Watch hours by genre

In [6]:

```
genre_hours = wh_titles.groupby('primary_genre')['watch_duration_min'].sum() / 60
genre_hours = genre_hours.sort_values()

genre_hours.plot(kind='barh', figsize=(9,6))
plt.title('Total Watch Hours by Genre')
plt.xlabel('Watch Hours')
plt.show()
```



In [7]:

```
print(genre_hours.sort_values(ascending=False).head(3))
```

```
primary_genre
Drama      501884.466667
Comedy     423149.053333
Action     393171.916667
Name: watch_duration_min, dtype: float64
```

Takeaway: Drama, Comedy, and Action are the top 3 genres by watch hours. The smallest genres (Musical, Stand-Up) get much less viewing time.

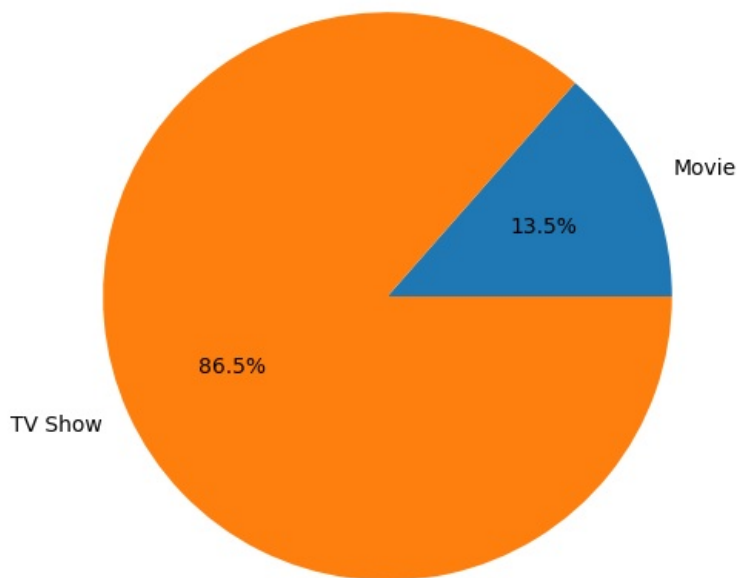
Chart 4: Movies vs TV Shows (share of watch hours)

In [8]:

```
type_hours = wh_titles.groupby('type')['watch_duration_min'].sum() / 60

plt.figure(figsize=(6,6))
plt.pie(type_hours, labels=type_hours.index, autopct='%1.1f%%')
plt.title('Watch Hours: Movies vs TV Shows')
plt.show()
print(type_hours)
```

Watch Hours: Movies vs TV Shows



```
type
Movie      4.508588e+05
TV Show    2.882609e+06
Name: watch_duration_min, dtype: float64
```

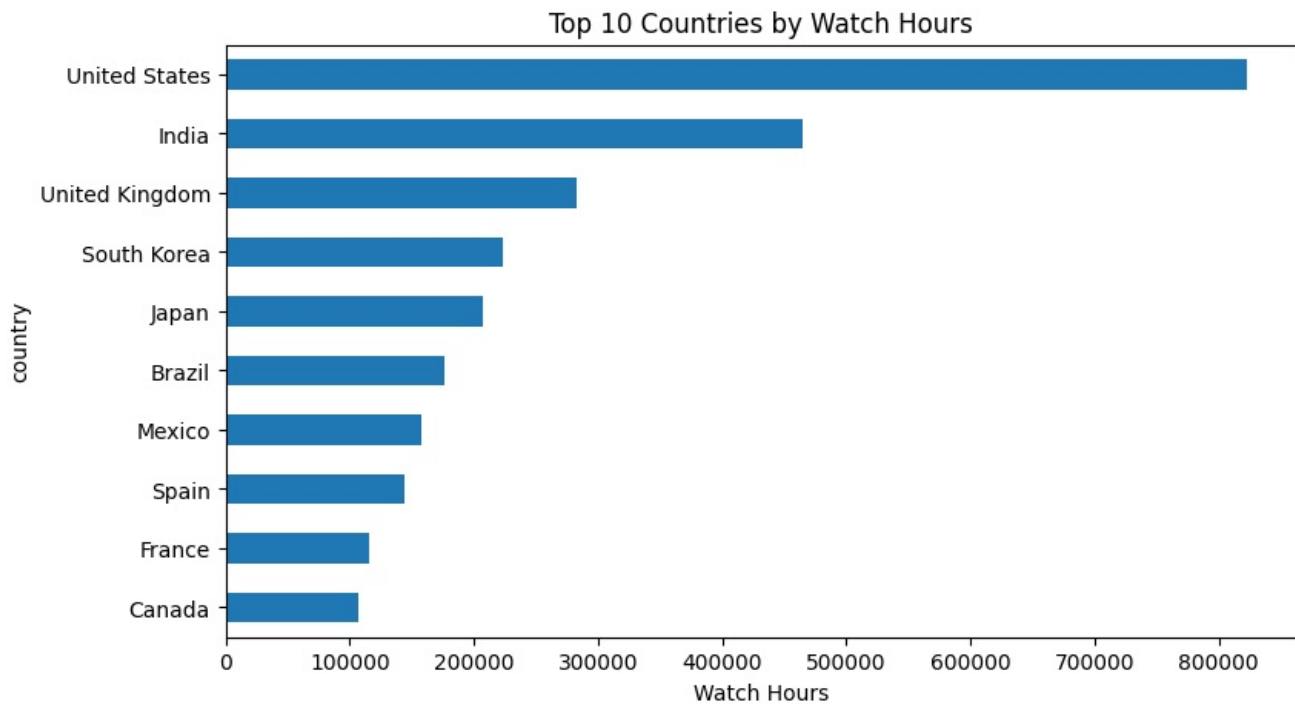
Observation: TV Shows account for more total watch hours than Movies, even though there are more Movie titles in the catalogue. This makes sense because a TV Show has many episodes a subscriber can binge, while a Movie is just one sitting.

Chart 5: Top 10 countries by watch hours

In [9]:

```
country_hours = wh_titles.groupby('country')['watch_duration_min'].sum() / 60
top10_countries = country_hours.sort_values(ascending=False).head(10)
top10_countries = top10_countries.sort_values()

top10_countries.plot(kind='barh', figsize=(9,5))
plt.title('Top 10 Countries by Watch Hours')
plt.xlabel('Watch Hours')
plt.show()
```



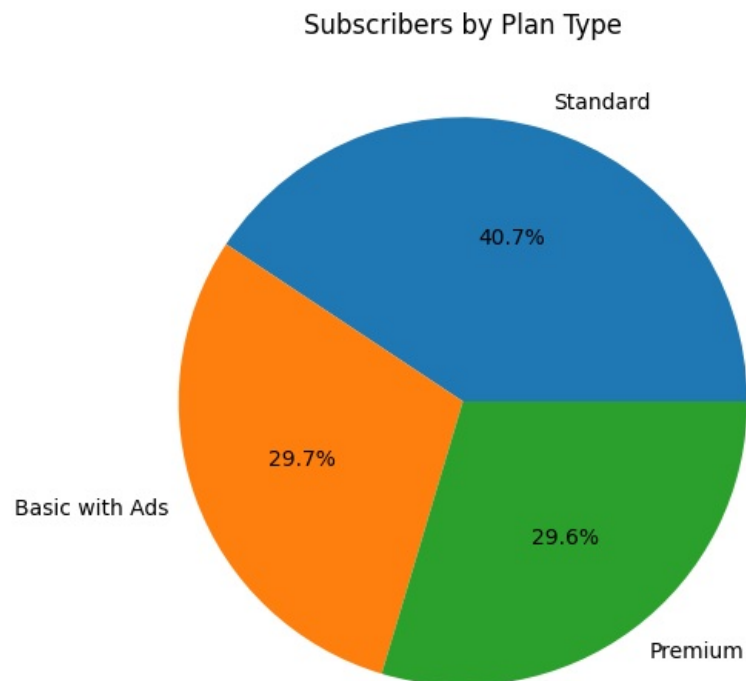
What stands out: The United States is clearly the biggest market. India, the UK, and South Korea are the next largest.

Chart 6: Subscriber plan distribution

In [10]:

```
plan_counts = subscribers['plan_type'].value_counts()

plt.figure(figsize=(6,6))
plt.pie(plan_counts, labels=plan_counts.index, autopct='%1.1f%%')
plt.title('Subscribers by Plan Type')
plt.show()
print(plan_counts)
```



```
plan_type
Standard      6100
Basic with Ads 4458
Premium       4442
Name: count, dtype: int64
```

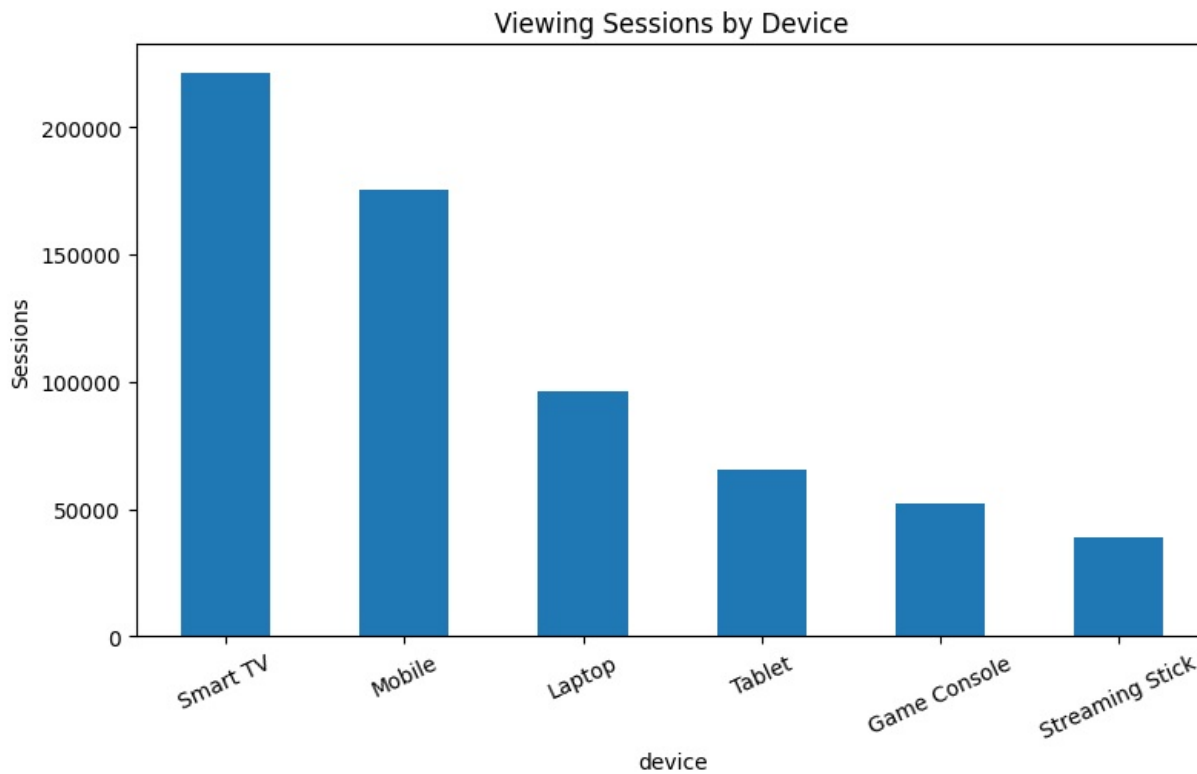
Observation: Standard is the most common plan. Basic with Ads and Premium are close to each other in size.

Chart 7: Device usage

In [11]:

```
device_counts = watch_history['device'].value_counts()

device_counts.plot(kind='bar', figsize=(9,5))
plt.title('Viewing Sessions by Device')
plt.ylabel('Sessions')
plt.xticks(rotation=25)
plt.show()
print(device_counts)
```



```
device
Smart TV      221446
Mobile        175605
Laptop         96408
Tablet         65276
Game Console   52322
Streaming Stick 38943
Name: count, dtype: int64
```

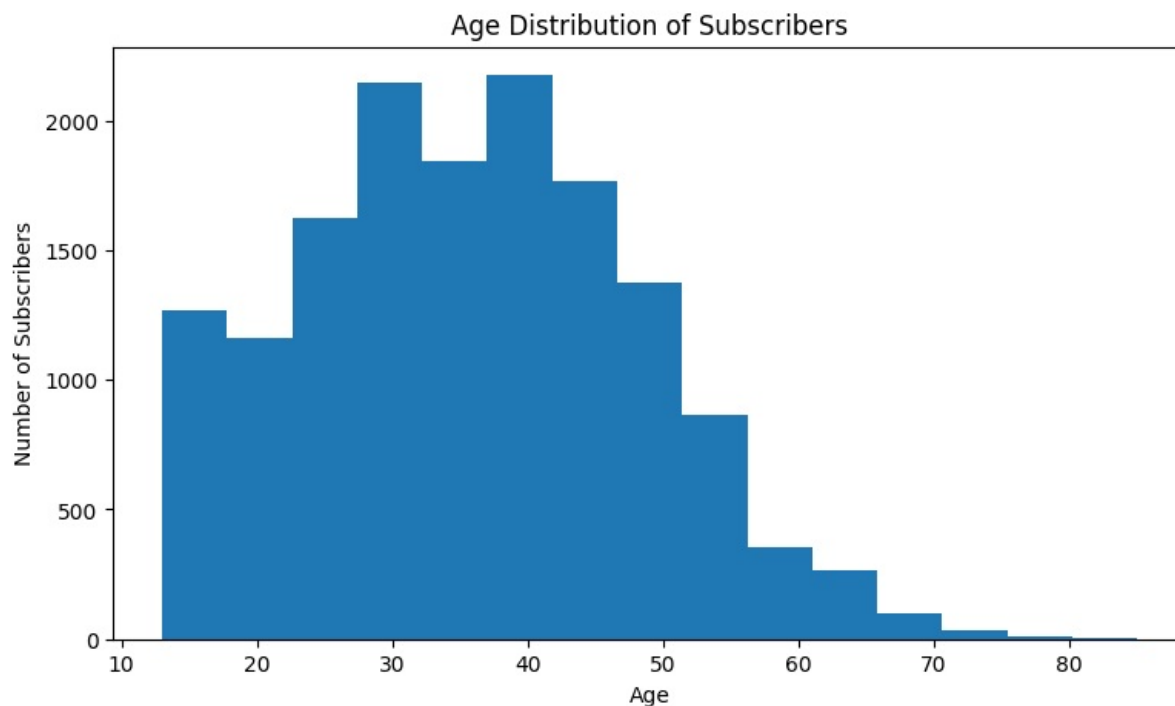
Note: Smart TV is the most used device, followed by Mobile. Together they make up more than half of all viewing sessions.

Chart 8: Age distribution of subscribers

In [12]:

```
plt.figure(figsize=(9,5))
plt.hist(subscribers['age'], bins=15)
plt.title('Age Distribution of Subscribers')
plt.xlabel('Age')
plt.ylabel('Number of Subscribers')
plt.show()

print(subscribers['age'].describe())
```



```
count    15000.000000
mean      35.620267
std       12.532555
min       13.000000
25%       27.000000
50%       35.000000
75%       44.000000
max       85.000000
Name: age, dtype: float64
```

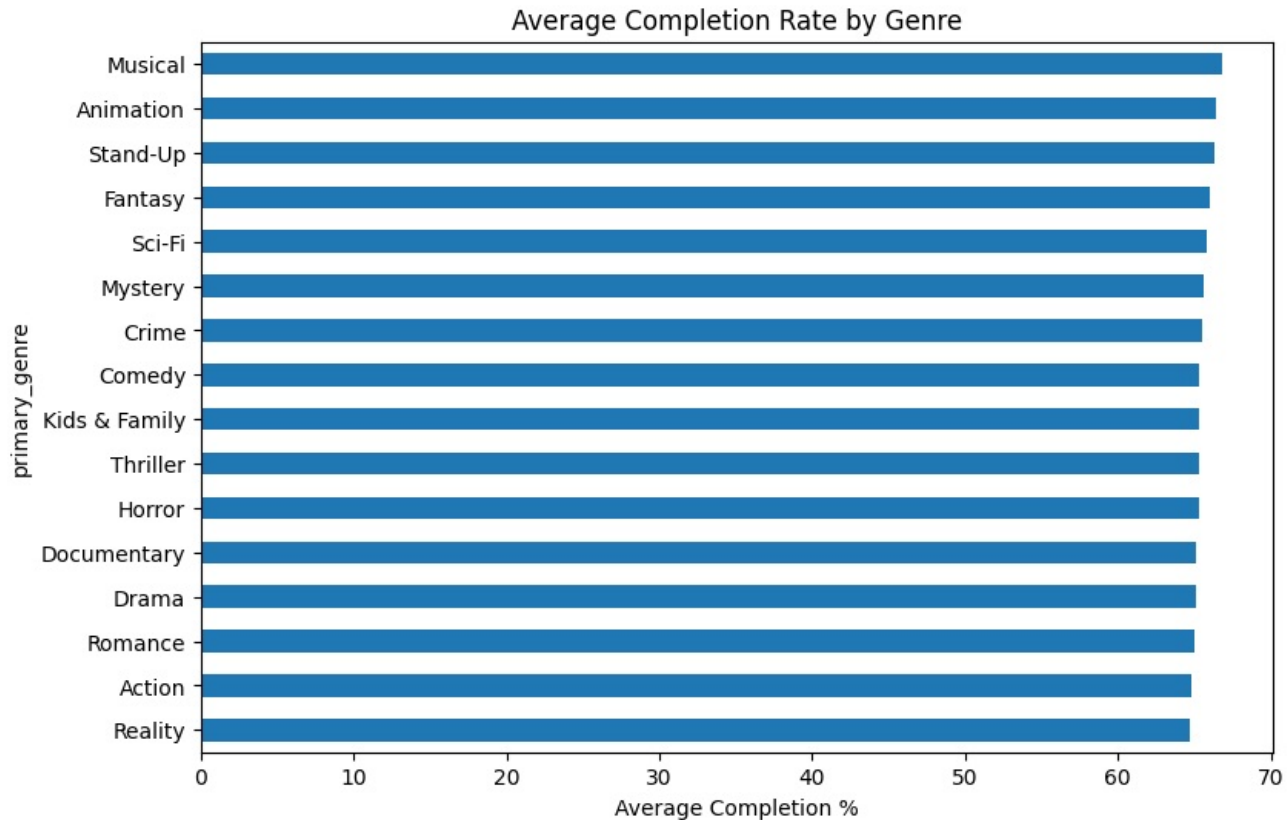
Observation: Most subscribers are between about 25 and 45 years old. There are fewer teenage and senior subscribers.

Chart 9: Completion rate by genre

In [13]:

```
completion_by_genre = wh_titles.groupby('primary_genre')['completion_pct'].mean()
completion_by_genre = completion_by_genre.sort_values()

completion_by_genre.plot(kind='barh', figsize=(9,6))
plt.title('Average Completion Rate by Genre')
plt.xlabel('Average Completion %')
plt.show()
```



In [14]:

```
print('Highest completion:')
print(completion_by_genre.sort_values(ascending=False).head(3))
print()
print('Lowest completion:')
print(completion_by_genre.sort_values().head(3))
```

```
Highest completion:
primary_genre
Musical      66.834788
Animation    66.383549
Stand-Up     66.319730
Name: completion_pct, dtype: float64
```

```
Lowest completion:
primary_genre
Reality      64.700084
Action       64.812501
Romance      65.004707
Name: completion_pct, dtype: float64
```

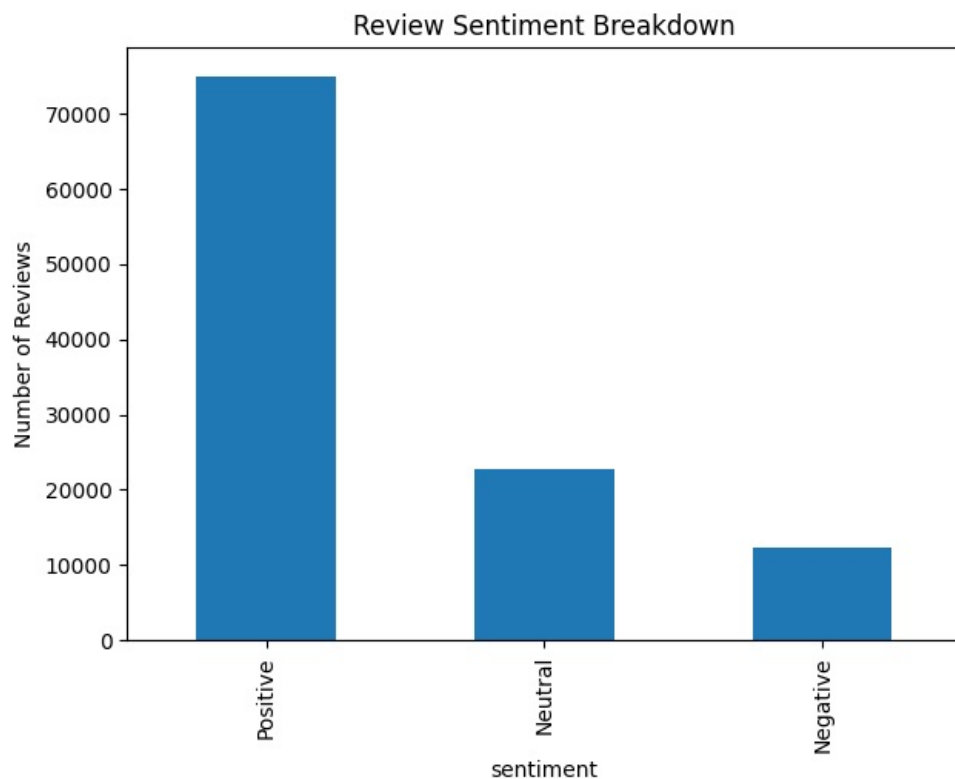
Takeaway: Completion rates are fairly close across genres, mostly between 62% and 68%. Documentary and Stand-Up finish a little higher, maybe because they are shorter and easier to finish in one sitting.

Chart 10: Review sentiment breakdown

In [15]:

```
sentiment_counts = reviews['sentiment'].value_counts()

sentiment_counts.plot(kind='bar', figsize=(7,5))
plt.title('Review Sentiment Breakdown')
plt.ylabel('Number of Reviews')
plt.show()
print(sentiment_counts)
```



```
sentiment
Positive    75007
Neutral     22759
Negative     12234
Name: count, dtype: int64
```

In [16]:

```
positive_share = sentiment_counts['Positive'] / sentiment_counts.sum() * 100
print('Percent of reviews that are positive:', round(positive_share, 1), '%')
```

Percent of reviews that are positive: 68.2 %

Observation: Most reviews are positive (around 68%). Negative reviews are the smallest group. Overall, subscribers seem to like what they're watching.